

Stereochemical Control of Redox Co^{II}/Co^{III}-Cages with Switchable Cotton Effects Based on Labile-Static States

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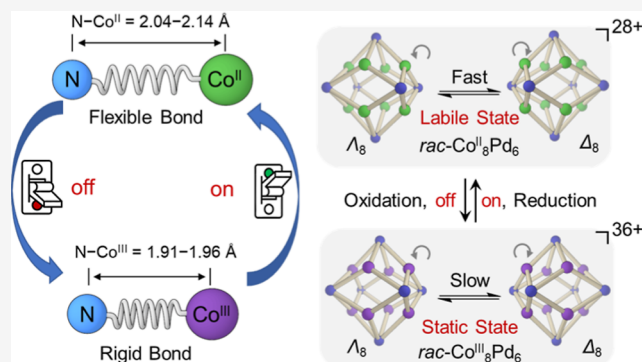
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ABSTRACT: The structural dynamics of artificial assemblies, in aspects such as molecular recognition and structural transformation, provide us with a blueprint to achieve bioinspired applications. Here, we describe the assembly of redox-switchable chiral metal-organic cages Λ_8/Δ_8 -[Pd₆(Co^{II}L₃)₈]²⁸⁺ and Λ_8/Δ_8 -[Pd₆(Co^{III}L₃)₈]³⁶⁺. These isomeric cages demonstrate an on-off chirality logic gate controlled by their chemical and stereostructural dynamics tunable through redox transitions between the labile Co^{II}-state and static Co^{III}-state with a distinct Cotton effect. The transition between different states is enabled by a reversible redox process and chiral recognition originating in the tris-chelate Co-centers. All cages in two states are thoroughly characterized by NMR, ESI-MS, CV, CD, and X-ray crystallographic analysis, which clarify their redox-switching behaviors upon chemical reduction/oxidation. The stereochemical lability of the Co^{II}-center endows the Λ_8/Δ_8 -Co^{II}-cages with efficient chiral-induction by enantiomeric guests, leading to enantiomeric isomerization to switch between Λ_8/Δ_8 -Co^{II}-cages, which can be stabilized by oxidation to their chemically inert forms of Λ_8/Δ_8 -Co^{III}-cages. Kinetic studies reveal that the isomerization rate of the Δ_8 -Co^{III}-cage is at least an order of magnitude slower than that of the Δ_8 -Co^{II}-cage even at an elevated temperature, while its activation energy is 16 kcal mol⁻¹ higher than that of the Co^{II}-cage.



INTRODUCTION

The past decades have seen tremendous advancements in supramolecular chemistry, especially coordination-driven self-assembly of cage receptors, also known as metal-organic cages/containers (MOCs).¹⁻⁵ Artificial cage structures possess well-defined shapes, sizes, cavities, and designable functionalities, showing prospects in supramolecular catalysis,⁶⁻²⁰ recognition, separation,²¹⁻²⁶ reactive compounds stabilization,²⁷⁻³⁰ biological applications,³¹⁻³⁹ molecular machines and devices,⁴⁰⁻⁴⁵ and so on. As an indispensable component of MOCs, metal centers play a crucial role in determining their functionalities and structural diversities. Flexible cages constructed from labile metal centers can exhibit smart response and adaptability to external inputs, such as guests, solvents, and anions, thereby able to achieve structural and functional diversity.^{16,46} Kinetically inert metal centers, on the other hand, can better maintain structural stability, not only achieving high shape persistence but also effectively preserving the stereo configurations of chiral-at-metal supramolecular structures.⁴⁷⁻⁴⁹ Incorporating both of these features in a single cage is immensely valuable for the controllable manipulation of cage configurations, guest recognition, and function modulation.^{50,51}

In biological systems, redox-sensitive residues can act as modulators to reversibly regulate biofunctions through the reaction with reactive oxygen species (ROS).⁵² The structural characteristic transformation of local redox-sensitive residues can propagate down the helices containing these residues, thus leading to significant changes in the overall stability, heterogeneity, and activity, thereby enabling the regulation of protein functions and the transmission of signals.⁵³⁻⁵⁶ For artificial assemblies, the transformation of structural characteristics induced by redox processes also represents a promising method for manipulating the cage dynamics in solution without modifying the structure or changing the coordination center.^{51,57-59}

Our previous work manifested that changes in the metal centers with different labilities could lead to alterations in the overall properties of MOCs in terms of guest recognition and chirality induction.^{46,60} However, achieving distinct dynamics

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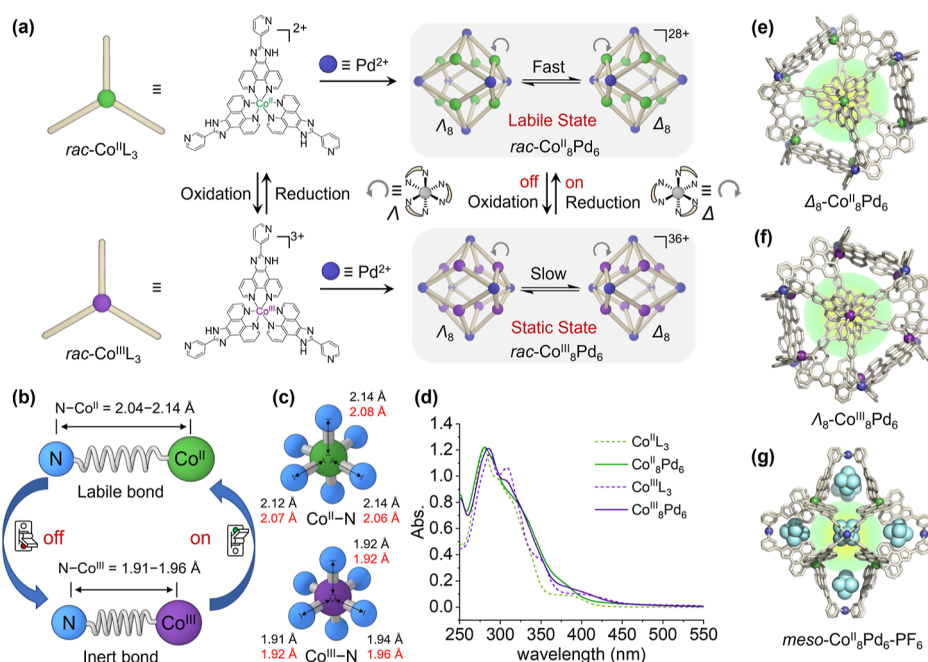


Figure 1. (a) Illustration of the synthetic routes and reversible redox conversion processes for rac-CoL_3 and $\text{rac-Co}_8\text{Pd}_6$ cage. (b) Schematic showing the distinct Co–N bond lengths and dynamics based on $\text{Co}^{\text{II}}/\text{Co}^{\text{III}}$ valence. (c) Local cobalt coordination environments and bond lengths in the crystal structures of $\text{rac-Co}^{\text{II}}_8\text{Pd}_6$ and $\text{rac-Co}^{\text{III}}_8\text{Pd}_6$ (red numbers), and $(S\text{-BINOL})_{24}\text{C}\Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and $(R\text{-BINOL})_{24}\text{C}\Lambda_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ (black numbers). (d) UV–vis spectra of $\text{rac-Co}^{\text{III}}_8\text{Pd}_6$, $\text{rac-Co}^{\text{II}}_8\text{Pd}_6$, $\text{rac-Co}^{\text{III}}\text{L}_3$, and $\text{rac-Co}^{\text{II}}\text{L}_3$ in CH_3CN . (e–g) Representative single-crystal structures of (e) $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ cage, (f) $\Lambda_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ cage, and (g) $\text{meso-Co}^{\text{II}}_8\text{Pd}_6\text{-PF}_6$ cage. Co^{II} : green ball; Co^{III} : violet ball; Pd^{II} : dark blue ball; N: sky blue; and organic ligand: wheat stick.

within an MOC by altering the metal centers involves rather elaborate synthetic steps.⁴³ Alternatively, redox-active metal centers, e.g., $\text{Co}^{\text{II}}/\text{Co}^{\text{III}}$ which can easily switch under mild conditions reversibly, thus becomes ideal building units in the assembly of dual-role MOC structures for in situ regulation (Figure 1a). Differentiated electronic configurations of $\text{Co}^{\text{II}}/\text{Co}^{\text{III}}$ -centers render distinct ligand field stabilization energies that affect the coordination bond lengths and strength (Figure 1b). The d^7 Co^{II} -centers in the lower charge state with longer Co–N bonds (>2.0 Å) tend to exhibit coordination dynamics and adopt a larger range of torsion angles; hence, it is widely applied in the regulation of flexible and responsive structures.^{16,61–63} In contrast, the d^6 Co^{III} -centers in the higher charge state generally show shorter bond lengths (<2.0 Å) and inert kinetics, less applied directly in the assembly of cage structures.^{51,64} Lusby group and Li group have respectively showcased the use of redox properties of $\text{Co}^{\text{II}}/\text{Co}^{\text{III}}$ -centers to obtain stable Co^{III} -cages as fixing controllers between smaller and larger cage structures.^{50,51,57} These excellent works demonstrated a convenient transformation pathway to construct highly stable Co^{III} -structures for configuration and static state fixation. However, utilization of such charge isomerism process to achieve controllable regulation of the physicochemical properties and optoelectronic switch is still underdeveloped, albeit with unique chiral merits.^{65,66}

In this work, we introduced redox-active $\text{Co}^{\text{II}}/\text{Co}^{\text{III}}$ -centers into triangular metalloligands CoL_3 and assembled with Pd^{II} to generate two pairs of chiral charge isomers of rhombic dodecahedron shaped cages, $\Lambda_8/\Delta_8\text{-}[\text{Pd}_6(\text{Co}^{\text{II}}\text{L}_3)_8](\text{BF}_4)_{28}$ and $\Lambda_8/\Delta_8\text{-}[\text{Pd}_6(\text{Co}^{\text{III}}\text{L}_3)_8](\text{BF}_4)_{36}$ [designated as MOC-16-Co(II) and MOC-16-Co(III) with $\text{Co}^{\text{II}}_8\text{Pd}_6$ and $\text{Co}^{\text{III}}_8\text{Pd}_6$ in short, Figure 1a]. The redox reactions of Co-centers could be

readily achieved under mild conditions, resulting in tuning of the cage charges between 28+ and 36+ while retaining the overall cage frameworks. Chiral $\Lambda_8/\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ cages were achievable through in situ synthesis, solution induction or crystallization process from stereolabile tris-chelate- $\text{Co}(\text{II})$ metalloligands, using enantiomeric *S/R*-BINOLs as chiral inducing guests.⁶⁷ Further in situ oxidation of Co^{II} to Co^{III} enhanced the structural stability of the cages, fixing their absolute configurations to prevent racemization even under chiral induction conditions. The kinetic investigation unveiled an increase of the activation energy from 9.43 ± 0.78 kcal mol^{-1} of Co^{II} -cage to 25.44 ± 1.44 kcal mol^{-1} of the oxidized Co^{III} -cage. Consequently, a reversible switch between $\Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ cages could be facilitated by chiral recognition of enantiomeric guests, while reversible transformation between their labile state and static state could be triggered by oxidant and reductant. These transformations could lead to a chiral on–off logic gate mediated by stereochemical dynamics of $\text{Co}^{\text{II/III}}$ -cages which exhibit a distinct and characteristic Cotton effects.

RESULTS AND DISCUSSION

Assembly of racemic Co_8Pd_6 Cages. The divalent $\text{Co}^{\text{II}}\text{L}_3$ metalloligand was synthesized by stirring $\text{Co}(\text{BF}_4)_2 \cdot 6\text{H}_2\text{O}$ and 2-(pyridin-3-yl)-1*H*-imidazo[4,5-*f*][1,10]-phenanthroline (L) in a DMSO solution at room temperature, which can be further oxidized into its trivalent form $\text{Co}^{\text{III}}\text{L}_3$ by H_2O_2 . Due to the effect of contact shift from unpaired electron, protons adjacent to the Co^{II} -center on $\text{Co}^{\text{II}}\text{L}_3$ downfield shifted to 17–100 ppm in the ^1H NMR spectrum (Figures S1–S6). Whereas the weak paramagnetic relaxation enhancement effect of the high-spin Co^{II} made its NMR signals narrower and clearer compared to other paramagnetic metal complexes, enabling

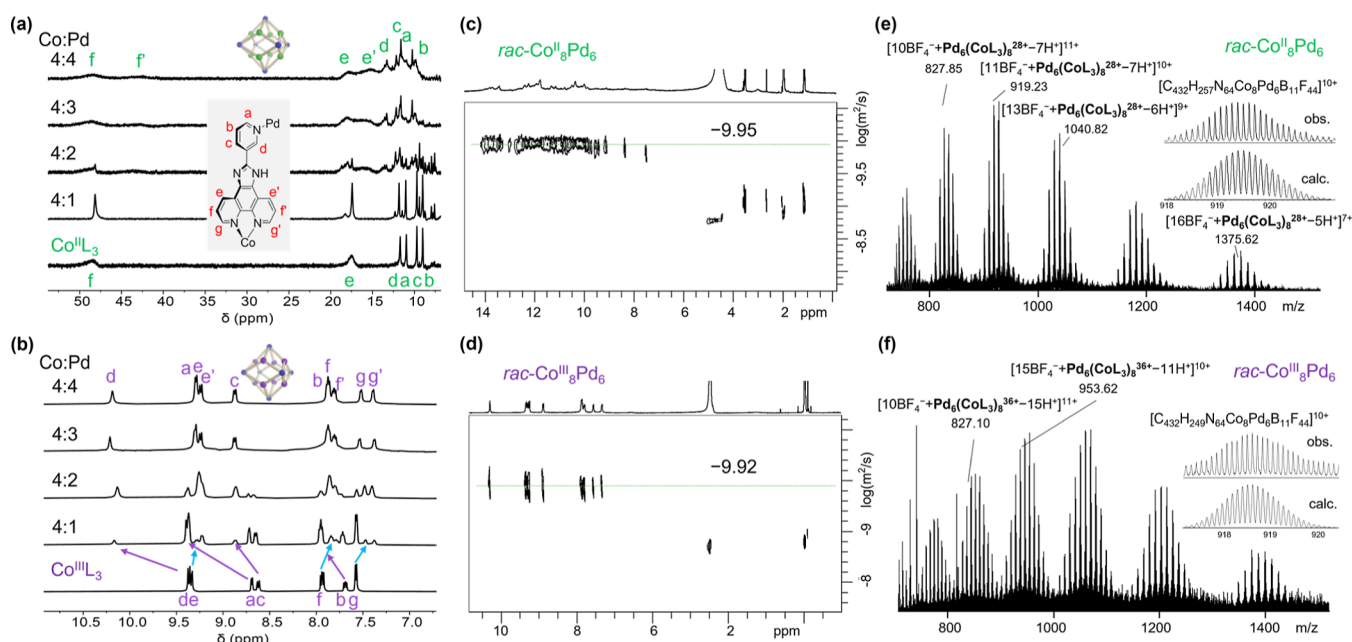


Figure 2. (a,b) ^1H NMR titration of (a) $\text{Co}^{\text{II}}\text{L}_3(\text{BF}_4)_2$ and (b) $\text{Co}^{\text{III}}\text{L}_3(\text{BF}_4)_3$ with Pd^{2+} in different ratios after heating at 60°C for 6 h. Shifts of the proton peaks in $\text{rac-Co}^{\text{II/III}}\text{Pd}_6$ are shown by arrows (400 MHz, $\text{CD}_3\text{CN}/\text{D}_2\text{O} = 1:4$, v/v, metalloligands in CD_3CN , 298 K). (c,d) ^1H DOSY NMR spectra of (c) $\text{rac-Co}^{\text{II}}\text{Pd}_6$ and (d) $\text{rac-Co}^{\text{III}}\text{Pd}_6$ cages (400 MHz, $\text{CD}_3\text{CN}/\text{D}_2\text{O} = 1:4$, v/v, 298 K). (e,f) HR-ESI mass spectra of (e) $\text{rac-Co}^{\text{II}}\text{Pd}_6$ and (f) $\text{rac-Co}^{\text{III}}\text{Pd}_6$ (CH_3CN , positive ion mode). Inserts show isotopic patterns of the experimental and simulation.

definite identification of individual peaks.^{68,69} Conversely, transition to diamagnetic low-spin Co^{III} -complex after oxidation resulted in NMR signals predominantly located within the normal aromatic region (Figures S7–S11). Therefore, the NMR resonances can be used as an indicator to distinguish the valence states of the metal centers of the Co-complexes to some extent.

Formation of $\text{rac}-[(\text{Co}^{\text{II}}\text{L}_3)_8\text{Pd}_6]^{28+}$ and $\text{rac}-[(\text{Co}^{\text{III}}\text{L}_3)_8\text{Pd}_6]^{36+}$ cages proceeded readily by heating a CH_3CN mixture of stoichiometric $\text{rac-Co}^{\text{II/III}}/\text{Co}^{\text{III}}\text{L}_3$ metalloligands and $\text{Pd}(\text{BF}_4)_2$ in an inert gas atmosphere at 60°C for 6 h (Figures 1a and S12–S25). The assembly processes of $\text{rac-Co}^{\text{II}}\text{Pd}_6$ and $\text{rac-Co}^{\text{III}}\text{Pd}_6$ were monitored by virtue of ^1H NMR titration in a $\text{CD}_3\text{CN}/\text{D}_2\text{O}$ mixture, revealing that the reactions converged at the ratio of 3:4 in both cases (Figure 2a,b). The solution characterization of the $\text{rac-Co}^{\text{III}}\text{Pd}_6$ cage was performed by ^1H – ^1H COSY and NOESY, from which protons were unambiguously assigned. Despite severe shifts, ^1H DOSY revealed a unified hydrodynamic radius of $\text{Co}^{\text{II}}\text{Pd}_6$ cage, closely comparable to the size of $\text{Co}^{\text{III}}\text{Pd}_6$, indicating similar cage structures in solution (Figures 2c,d, S15, and S23). The $(\text{CoL}_3)_8\text{Pd}_6$ composition of cages were confirmed by high-resolution electrospray ionization-mass spectrometry (HR-ESI-MS), which disclosed the formula of a series of $\{[(\text{CoL}_3)_8\text{Pd}_6]^{28+} - x\text{H}^+ + y(\text{BF}_4^-)\}^{(28-x-y)+}$ species for the $\text{rac-Co}^{\text{II}}\text{Pd}_6$ (Figure 2e) and $\{[(\text{CoL}_3)_8\text{Pd}_6]^{36+} - x\text{H}^+ + y(\text{BF}_4^-)\}^{(36-x-y)+}$ for the $\text{rac-Co}^{\text{III}}\text{Pd}_6$ (Figure 2f). Due to the tautomeric nature of the exchangeable proton on imidazole motifs, a series of proton–anion balanced signals appear in the same charged cage species.⁷⁰ Because the overall charge states of these two cages are +28 and +36, the species of $\text{Co}^{\text{III}}\text{Pd}_6$ cage necessitates additional loss of eight protons to reach the same charge of corresponding $\text{Co}^{\text{II}}\text{Pd}_6$ cage species if equivalent numbers of anion are bound. As a result, the discrepancy in molecular weights corresponding to m/z peaks

carrying the same charges between these two cages is approximately 8 Da (Figure S24).

UV–vis absorption spectra of $\text{rac-Co}^{\text{II}}\text{L}_3$, $\text{rac-Co}^{\text{III}}\text{L}_3$, $\text{rac-Co}^{\text{II}}\text{Pd}_6$, and $\text{rac-Co}^{\text{III}}\text{Pd}_6$ in CH_3CN were measured. The main peaks corresponding to the π – π^* transition of the organic ligand below 300 nm were slightly red-shifted in $\text{Co}^{\text{III}}\text{L}_3$ and $\text{Co}^{\text{III}}\text{Pd}_6$ compared to $\text{Co}^{\text{II}}\text{L}_3$ and $\text{Co}^{\text{II}}\text{Pd}_6$. However, significant changes in intensity and position of the bands in 300–450 nm region were observed between $\text{Co}^{\text{III}}\text{L}_3$, $\text{Co}^{\text{II}}\text{Pd}_6$ and $\text{Co}^{\text{II}}\text{L}_3$, $\text{Co}^{\text{III}}\text{Pd}_6$, which were contributed by n – π^* transitions as well as the metal-to-ligand charge-transfer (MLCT) arisen from Co^{II} -centers and the ligand-to-metal charge-transfer (LMCT) arisen from Co^{III} -centers^{71,72} (Figure 1d). Oxidation of the divalent Co^{II} -centers leads to expanded absorption in the visible range, accounting for the deeper color of the trivalent counterparts. It is noted that although the absorption profiles are generally similar for $\text{Co}^{\text{II}}\text{L}_3$ metalloligand and $\text{Co}^{\text{II}}\text{Pd}_6$ cage of the divalent Co^{II} -centers, or $\text{Co}^{\text{III}}\text{L}_3$ metalloligand and $\text{Co}^{\text{III}}\text{Pd}_6$ cage of the trivalent Co^{III} -centers, distinguishable intensity of the absorption bands are found between the corresponding metalloligands and cages, due to Pd^{II} -coordination to pyridine which brings in strong electron withdrawing effect, as well as Pd^{II} –pyridine MLCT,⁷³ after the cage formation.

Crystal Structures of Co_8Pd_6 Cage Isomers. The cage configurations of $\text{Co}^{\text{II}}\text{Pd}_6$ and $\text{Co}^{\text{III}}\text{Pd}_6$ were unambiguously confirmed by single-crystal X-ray diffraction analyses in the solid-state (Figures S30–S32, S37, and S38 and Tables S1 and S2). Suitable single-crystals for the in-house diffractometer were obtained by slow vapor diffusion of MeOH into a solution of $\text{rac-Co}^{\text{II}}\text{Pd}_6$ or $\text{rac-Co}^{\text{III}}\text{Pd}_6$ in CH_3CN . These cages crystallized in the same packing mode, and formed octahedral-shaped single-crystals in centrosymmetric tetragonal space group $P4/n$ with the asymmetric unit containing one-fourth of the whole cage. The differentiated bond lengths of Co^{II} –N (average 2.04 Å) and Co^{III} –N (average 1.94 Å) in rac –

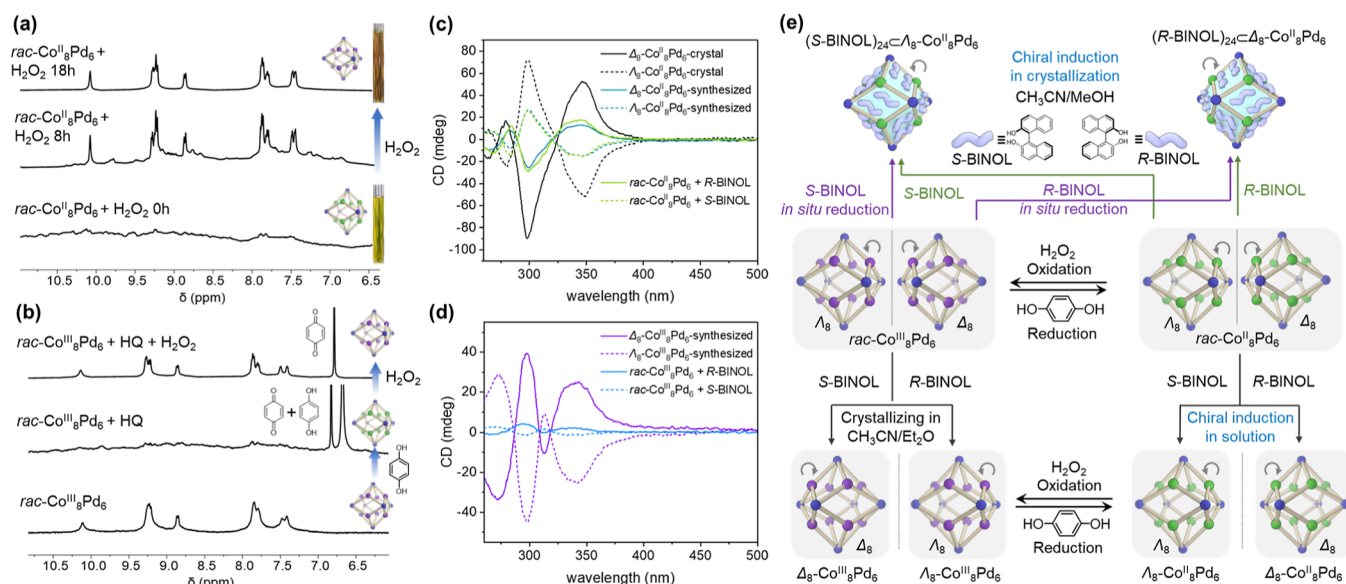


Figure 3. (a,b) Partial ^1H NMR spectra of (a) $\text{rac-Co}^{\text{II}}_8\text{Pd}_6$ with the addition of $5\ \mu\text{L}$ 30 wt % H_2O_2 at room temperature for varied times, and (b) $\text{rac-Co}^{\text{III}}_8\text{Pd}_6$ after the addition of 20 equiv HQ and then $5\ \mu\text{L}$ of 30 wt % H_2O_2 (400 MHz, $\text{CD}_3\text{CN}/\text{D}_2\text{O} = 1:4$, v/v, 298 K). (c) CD spectra of the dissolved chiral Δ_8 - and Λ_8 - $\text{Co}^{\text{II}}_8\text{Pd}_6$ crystals, in situ synthesized Δ_8 - and Λ_8 - $\text{Co}^{\text{II}}_8\text{Pd}_6$ and the direct induction of $\text{rac-Co}^{\text{II}}_8\text{Pd}_6$ with R- or S-BINOL (R/S-BINOL inducers were removed before CD measurement, 300 K, 1.6×10^{-6} M, CH_3CN). (d) CD spectral comparison of the synthesized Δ_8 - and Λ_8 - $\text{Co}^{\text{III}}_8\text{Pd}_6$ from oxidation of chiral Δ_8/Λ_8 - $\text{Co}^{\text{II}}_8\text{Pd}_6$, and the direct induction of $\text{rac-Co}^{\text{III}}_8\text{Pd}_6$ with R- or S-BINOL (R/S-BINOL inducers were removed before CD measurement, 1.6×10^{-6} M, CH_3CN). (e) Illustration of the chiral induction processes in crystallization or in solution, in combination with the reversible redox processes.

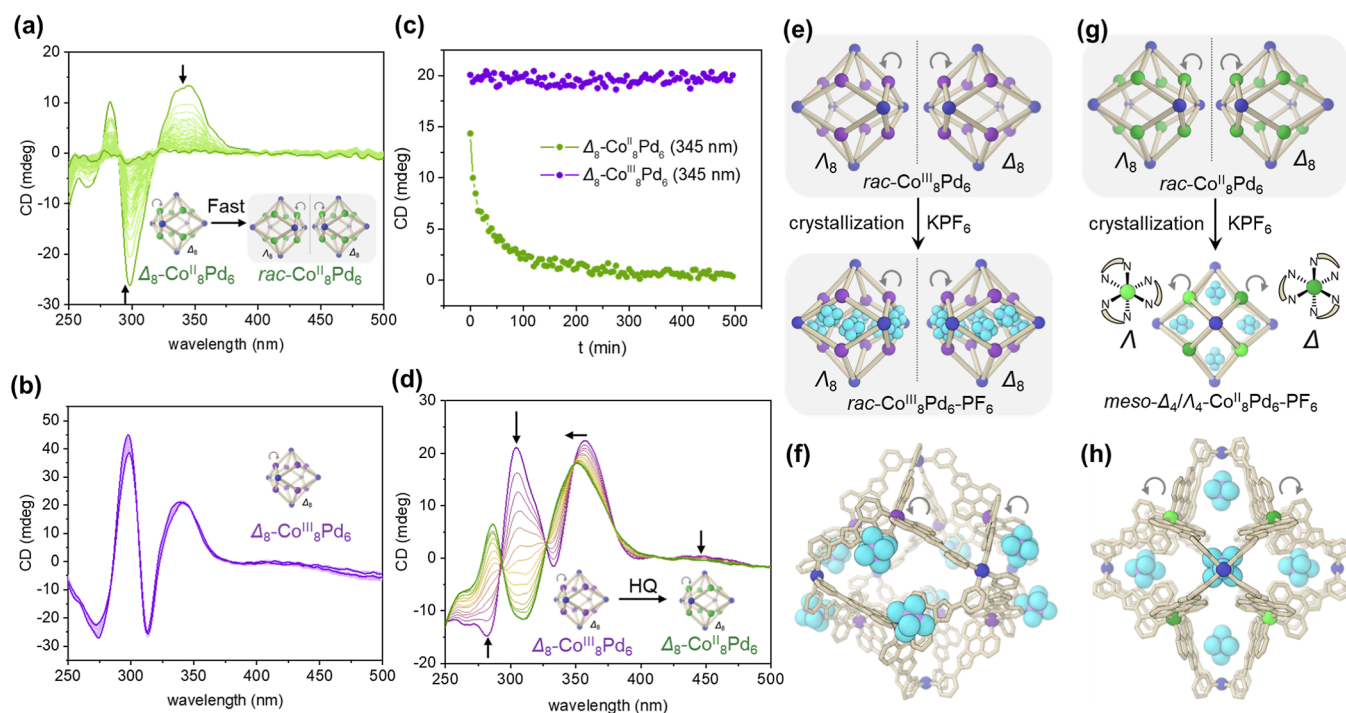


Figure 4. (a,b) CD signal changes of (a) $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and (b) $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ after dissolving in CH_3CN for 8 h (298 K, 1.6×10^{-6} M). (c) CD signal change profiles for $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ (green) and $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ (purple) monitored at 345 nm (298 K, 1.6×10^{-6} M, CH_3CN). (d) CD profile changes from $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ to $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ in the presence of 100 equiv HQ as reductant (308 K, 1.6×10^{-6} M, $\text{CH}_3\text{CN}/\text{H}_2\text{O} = 1:4$, v/v). (e,g) Illustration of the PF_6^- anion induced structural transformation for (e) $\text{rac-Co}^{\text{III}}_8\text{Pd}_6$ and (g) $\text{rac-Co}^{\text{II}}_8\text{Pd}_6$. (f,h) Crystal structures of (f) $\text{rac-Co}^{\text{III}}_8\text{Pd}_6\text{-PF}_6$ (only $\Lambda_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ presented) and (h) $\text{meso-}\Delta_4/\Lambda_4\text{-Co}^{\text{II}}_8\text{Pd}_6\text{-PF}_6$. Hydrogen atoms and other anions are omitted for clarity; $\Lambda\text{-Co}^{\text{II}}$ and $\Delta\text{-Co}^{\text{II}}$ centers are present with light green balls and dark green balls, respectively, in the $\text{meso-}\Delta_4/\Lambda_4\text{-Co}^{\text{II}}_8\text{Pd}_6\text{-PF}_6$ structure.

$\text{Co}^{\text{II}}_8\text{Pd}_6$ and $\text{rac-Co}^{\text{III}}_8\text{Pd}_6$ (Figure 1e,f and Table S1)⁵⁰ reflect the distinct valence states of the Co-centers in these two cages. The bond lengths of $\text{Co}^{\text{III}}\text{-N}$ align well with those typical low-spin d^6 Co^{III} -centers.⁷⁴ Notably, the $\text{rac-Co}^{\text{II}}_8\text{Pd}_6$ exhibits

shorter bond lengths compared to typical high-spin d^7 Co^{II} -centers, but no remarkable $\text{Co}^{\text{II}}\text{-N}$ bond variation from Jahn–Teller distortion is observed, suggesting that Co^{II} -centers are of high-spin d^7 configuration. The shortened bond lengths may

stem from the partial oxidation of Co^{II} -centers during the long-term crystallization process, leading to a decrease in the average bond lengths. However, when a reducing condition was applied for the chiral cage crystal growth in the presence of *R/S*-BINOL, the Co^{II} -N bond lengths become well comparable with those typically observed in high-spin d^7 Co^{II} -centers (Figure 1c).

Due to the strong stereocoupling effect between metal-oligands, these M_8Pd_6 -type cages tend to assemble in a homochiral fashion, which are considered thermodynamically stable isomers.^{49,67} Therefore, both Co^{II} -cage and Co^{III} -cage form a pair of homochiral $\Lambda_8/\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and $\Lambda_8/\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ enantiomers, respectively, which coexist in a single-crystal to give rise to racemates. This observation suggested that the cage assembly proceeds through chiral recognition between Λ - and Δ - $\text{Co}^{\text{II/III}}\text{L}_3$ metalloligands to self-sort homochiral ones for cage construction, leading to the simultaneous formation of $\Lambda_8\text{-Co}_8\text{Pd}_6$ and $\Delta_8\text{-Co}_8\text{Pd}_6$ cages no matter starting from $\Lambda/\Delta\text{-Co}^{\text{II}}\text{L}_3$ or $\Lambda/\Delta\text{-Co}^{\text{III}}\text{L}_3$ metalloligands.^{49,67} To obtain enantiopure cages, syntheses were carried out in the presence of excess *R*- or *S*-BINOL for chiral induction.^{46,67} As expected, $\Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$, or $\Lambda_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ and $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ enantiomeric cages, were successfully achieved from *rac*- $\text{Co}^{\text{II}}\text{L}_3$ metalloligands when reactions were protected in a N_2 atmosphere or in situ oxidized by H_2O_2 , respectively (Figures 3c,d and S26–S29).

^1H NMR analyses of the dissolved crystals of *rac*- $\text{Co}^{\text{II}}_8\text{Pd}_6$ and *rac*- $\text{Co}^{\text{III}}_8\text{Pd}_6$ cages confirmed that distinct valence states of Co-centers were maintained in solution to survive ambient conditions (Figures S33 and S34). The NMR resonances of both crystalline sample and as-synthesized sample of *rac*- $\text{Co}^{\text{III}}_8\text{Pd}_6$ are almost the same and clean, suggesting neat structures in both solution and solid states, whereas the as-prepared *rac*- $\text{Co}^{\text{II}}_8\text{Pd}_6$ cage displayed some minor peaks compared with the crystalline sample, implying the existence of kinetic isomers. The addition of BINOL guests into the *rac*- $\text{Co}^{\text{II}}_8\text{Pd}_6$ cage solution showed little improvement of the cage resonances (Figure S39); however, the oxidation process could transform minor isomers to the thermodynamically stable homochiral cages (Figure 3a), giving unified and well-resolved proton signals of the oxidized Co^{III} cage.

As we previously reported for the dynamic $\text{Zn}^{\text{II}}_8\text{Pd}_6\text{-MOC-16}$ cage,⁴⁶ counteranions play an important role to induce the cage mesomerization process. In the presence of excess PF_6^- during the crystal growth of the *rac*- $\text{Co}^{\text{II}}_8\text{Pd}_6$ cage, we obtained a diastereomer composed of heterochiral metalloligands, forming a $\Delta\Delta\Delta\Delta\Delta\Delta\Delta\Delta\text{-meso-Co}^{\text{II}}_8\text{Pd}_6\text{-PF}_6$ cage (Figures 1g and 4g,h). In contrast to *rac*- $\text{Co}^{\text{II}}_8\text{Pd}_6$ comprising equivalent $\Lambda_8/\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ cages with each cage consisting of eight homochiral Λ - or Δ - $\text{Co}^{\text{II}}\text{L}_3$ metalloligands, the *meso*- $\text{Co}^{\text{II}}_8\text{Pd}_6\text{-PF}_6$ cage itself contains four $\Lambda\text{-Co}^{\text{II}}\text{L}_3$ and four $\Delta\text{-Co}^{\text{II}}\text{L}_3$ metalloligands with each of the 12 pockets consisting of two $\Lambda/\Delta\text{-Co}^{\text{II}}\text{L}_3$ metalloligands in opposite stereo configurations (Figures S30 and S31). Unfortunately, the paramagnetism of the Co^{II} -centers in the *meso*- $\text{Co}^{\text{II}}_8\text{Pd}_6\text{-PF}_6$ cage makes the protonal signals severely broadened, which prevented us from studying the solution structure explicitly (Figure S35). ^{19}F NMR spectrum of *meso*- $\text{Co}^{\text{II}}_8\text{Pd}_6$ also could not distinguish the PF_6^- anions inside or outside of the cage due to fast exchange (Figure S36). Nevertheless, single-crystal structural analysis evidently revealed multiple electrostatic and anion- π interactions in interior, exterior, and pocket sites between the cage and PF_6^- anions, which all contribute to the induction and

stabilization of the *meso*-cage⁴⁶ (Figure S32). Such anion-induced mesomerization⁴⁶ was not observed for $\text{Co}^{\text{III}}_8\text{Pd}_6\text{-PF}_6$ cages, indicative of significant difference in stereochemical dynamics between $\text{Co}^{\text{II}}_8\text{Pd}_6$ and $\text{Co}^{\text{III}}_8\text{Pd}_6$ cages (Figure 4e,f).

Redox Switch and Stability. Owing to the good redox behaviors, polypyridine Co-complexes have been widely applied as mediators in batteries and solar cells.^{74–76} The electrochemical properties of the $\text{Co}^{\text{II}}\text{L}_3$ metalloligand and *rac*- $\text{Co}^{\text{II}}_8\text{Pd}_6$ cage were investigated by cyclic voltammetry (CV) in CH_3CN (Figures S40 and S41), which showed two quasi-reversible redox potentials at $E_{1/2} = -1.30$ and -1.29 V, and $E_{1/2} = -0.04$ and -0.01 V (vs Fc), attributable to the $\text{Co}^{\text{II}}/\text{Co}^{\text{I}}$ and $\text{Co}^{\text{III}}/\text{Co}^{\text{II}}$ couples, respectively, in consistence with the known polypyridine Co-complexes.⁷⁴ Due to the long spatial $\text{Co}\cdots\text{Co}$ distances without conjugation linkers, the redox property of the Co-centers in the cage are not significantly affected by each other, leading to observation of the similar redox potentials for the cage and metalloligand. Such an independent redox process was also observed in other redox active coordination cages.^{64,70} These results suggest that the reversible redox transformations between $\text{Co}^{\text{II}}_8\text{Pd}_6$ and $\text{Co}^{\text{III}}_8\text{Pd}_6$ cages are implementable by selecting a suitable reductant and oxidant.

After careful screening of oxidants and reductants appropriate for reversible redox processes, we selected the simple hydrogen peroxide (H_2O_2) as an oxidant, which can smoothly achieve the oxidation of divalent metalloligands and the $\text{Co}^{\text{II}}_8\text{Pd}_6$ cage under mild conditions (Figures 3a and S42–S45). After adding excess H_2O_2 to the solution of *rac*- $\text{Co}^{\text{II}}_8\text{Pd}_6$ for 18 h, the initial paramagnetically broadened NMR resonances converted into a set of clear and well-resolved signals, consistent with the spectrum of *rac*- $\text{Co}^{\text{III}}_8\text{Pd}_6$, indicating complete oxidation of the Co^{II} cage to the Co^{III} cage. The color change of cage solution from light yellow to light brown also confirmed the valence transition process. The reduction of *rac*- $\text{Co}^{\text{III}}_8\text{Pd}_6$ is challenging due to possibly simultaneous reduction of $[\text{Pd}(\text{Py})_4]^{2+}$ moieties in six cage vertices, which may lead to the disassembly of the cage structure. After careful screening, hydroquinone (HQ) proved to be an ideal reductant that underwent oxidation under mild conditions to result in benzoquinone without interference in the cage structure (Figures 3b and S46–S50). HQ effectively reduced $\text{Co}^{\text{III}}_8\text{Pd}_6$ in $\text{CD}_3\text{CN}/\text{D}_2\text{O}$ (1/4, v/v), leading to the disappearance of the sharp resonances and appearance of broad paramagnetic signals in the range of 45–50 ppm, which are almost identical with the signals of *rac*- $\text{Co}^{\text{II}}_8\text{Pd}_6$. To further confirm the cage integrity during this reduction process, H_2O_2 was successively added to the above solution, leading to the oxidation of the Co^{II} -centers back to the trivalent form without structural dissociation. For comparison, the $\text{Fe}^{\text{II}}_8\text{Pd}_6$ and $\text{Ni}^{\text{II}}_8\text{Pd}_6$ isomorphous cages could not undergo similar redox transformation under such mild redox conditions due to the higher redox potentials of the polypyridyl Fe^{II} - and Ni^{II} -centers,^{70,77} highlighting the advantage of Co-centers in achieving valence transitions under mild conditions (Figures S51 and S52). Long-term stability tests showed that oxygen can also act as an oxidant to slowly oxidize $\text{Co}^{\text{II}}_8\text{Pd}_6$ partially into $\text{Co}^{\text{III}}_8\text{Pd}_6$ after its solution was placed in the air for 2 months. In contrast, no obvious change of $\text{Co}^{\text{III}}_8\text{Pd}_6$ was observed even after 7 months of keeping its solution in air (Figures S53 and S54).

The stronger trivalent $\text{Co}^{\text{III}}\text{-N}$ bond is known to be kinetically inert than the weaker divalent $\text{Co}^{\text{II}}\text{-N}$ bond that is

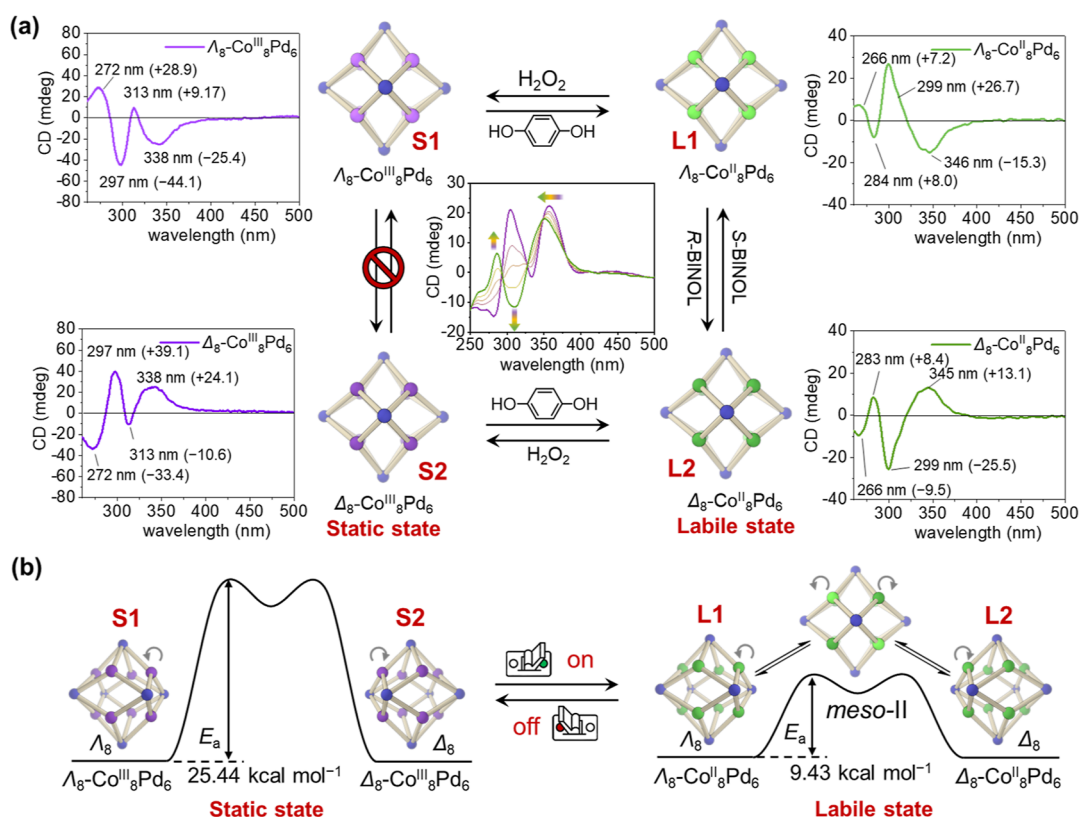


Figure 5. (a,b) Demonstration of a logic programming from the static S1 state of $\Lambda_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ to S2 state of $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ via redox transitions between Co^{III} and Co^{II} cages, and chirality conversions between the L1 state of $\Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and L2 state of $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ in the labile forms, and vice versa. The characteristic CD patterns corresponding to each individual state are depicted with main signals marked for comparison. The central inset showcasing CD signal variation from $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ to $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$. $\Lambda\text{-Co}^{\text{III}}$: light violet ball; $\Delta\text{-Co}^{\text{III}}$: dark violet ball; $\Lambda\text{-Co}^{\text{II}}$: light green ball; and $\Delta\text{-Co}^{\text{II}}$: dark green ball. (b) Diagram illustrating stereo configurational transformations between $\Delta_8\text{-Co}_8\text{Pd}_6$ and $\Lambda_8\text{-Co}_8\text{Pd}_6$ cages in different valent states, and the apparent activation energy E_a of $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ and $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ that turn-on and turn-off the structural conversions.

labile for structural and conformational changes,^{50,51,63–65} which remarkably impacts the chemical stability and stereochemical dynamics of the cage after redox switch. The $\text{Co}^{\text{II/III}}\text{-N}$ bonds stability was estimated using the divalent and trivalent $\text{Co}^{\text{II/III}}\text{L}_3$ metalloligands under acidic conditions (Figures S57 and S58). Upon addition of 1 M D_2SO_4 , the paramagnetic signals of $\text{Co}^{\text{II}}\text{L}_3$ disappeared and the free L signals emerged, indicative of the lability of the $\text{Co}^{\text{II}}\text{-N}$ bond. In contrast, $\text{Co}^{\text{III}}\text{L}_3$ exhibited stronger resistance to acid, and no dissociation was observed even in the presence of 1 M D_2SO_4 . Whereupon, the $\text{rac-Co}^{\text{III}}_8\text{Pd}_6$ cage was tested to be more stable than the reduced $\text{rac-Co}^{\text{II}}_8\text{Pd}_6$ cage (Figures S59 and S60). The $\text{rac-Co}^{\text{II}}_8\text{Pd}_6$ cage completely dissociated in a 1 M D_2SO_4 solution, whereas $\text{rac-Co}^{\text{III}}_8\text{Pd}_6$ only partially dissociated which was mostly caused by the dissociation of Pd-centers rather than $\text{Co}^{\text{III}}\text{-N}$ bonds. Therefore, the $\text{Co}^{\text{II}}_8\text{Pd}_6$ cage presents a labile state to facilitate stereochemical transformation inducible via chiral recognition,^{46,60} while its oxidized state $\text{Co}^{\text{III}}_8\text{Pd}_6$ is stable and static, enabling locking of the cage configuration.⁵⁰ This redox process holds a promise to serve as a switch to turn on or turn off the dynamics of the Co_8Pd_6 cage for a functionalization purpose (Figures 1a and 3e).

Chiral Induction and Stereochemical Logic Control.

On the basis of above valence determined characteristic labile and static states transition of the cage, we wondered whether the locking and unlocking of the cage dynamics could be applied to chirality control for enantiomeric guest recognition

and Cotton effect modulation through the redox processes (Figure 3c–e). As we reported before,⁴⁶ chiral guests induced cocrystallization readily enabled resolution and transformation of the racemic cages to afford 24 BINOL enclathrated ($S\text{-BINOL}$)₂₄ $\subset$ $\Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ or ($R\text{-BINOL}$)₂₄ $\subset$ $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ (Figures 3e and S37 and Table S2), benefited from the labile $\text{Co}^{\text{II}}\text{-N}$ bonds which permit configurational inversion between Λ_8 - and $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ cages.^{46,60} However, this method was not applicable to $\Lambda_8/\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ cage transformation due to the inert nature of $\text{Co}^{\text{III}}\text{-N}$ bonds. The attempts to resolve chiral Co^{III} -cages using $S/R\text{-BINOLs}$ led to the partial reduction of $\Lambda_8/\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ probably due to the oxidation of BINOL,⁷⁸ which has been testified by heating $R\text{-BINOL}$ and $\text{rac-Co}^{\text{III}}_8\text{Pd}_6$ in a $\text{CD}_3\text{CN}/\text{CD}_3\text{OD}$ mixture at 70 °C (Figure S55). Nevertheless, this reduction process becomes ineffective in aprotic acetonitrile media (Figure S56). Therefore, the enantiopure $\Lambda_8/\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ cages could be either obtained through in situ oxidation of the resolved $\Lambda_8/\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ with H_2O_2 (Figure S28), or through slow crystallization in the aprotic $\text{CH}_3\text{CN}/\text{Et}_2\text{O}$ mixture. Similar with our previously reported results, the solvophobic effects significantly affected the chiral recognition behaviors due to the solubility difference of BINOL guests and the enclathration ability of the cage pockets in different solvents.^{46,67} In aprotic $\text{CH}_3\text{CN}/\text{Et}_2\text{O}$ media, the single-guest binding mode was formed to give rise to a host–guest pair of ($R\text{-BINOL}$)₈ $\subset$ $\Lambda_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ or ($S\text{-BINOL}$)₈ $\subset$ $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ (Figure S38), which showed opposite chiral recognition behavior as compared to the

double-guest binding mode in $(S\text{-BINOL})_{24} \subset \Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ or $(R\text{-BINOL})_{24} \subset \Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ in a $\text{CH}_3\text{CN}/\text{CH}_3\text{OH}$ system.⁴⁶

We previously found that the chiral induction could be realized for in situ assembly of labile $\Lambda_8/\Delta_8\text{-Zn}_8\text{Pd}_6$ and $\Lambda_8/\Delta_8\text{-Fe}_8\text{Pd}_6$ cages, but ineffective for the kinetically inert Ni_8Pd_6 and Ru_8Pd_6 cages.^{46,67} This proved to be feasible to control the chirality of the stable Co^{III} -cages, taking advantage of the lability difference and valence transition between Co^{II} - and Co^{III} -cages to achieve in situ chiral induced switch of $\Lambda_8/\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ cages (Figure 3e), which themselves were difficult to realize from a direct conversion method. As expected, the chiral induction of $\text{rac-Co}^{\text{II}}_8\text{Pd}_6$ in the presence of excess *R*- or *S*-BINOL inducers resulted in conversion into enantioexcess Δ_8 - or $\Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$, respectively, as confirmed by the circular dichroism (CD) spectral measurements after removing the BINOLs (Figure 3c,e). Compared with the CD signals of dissolved chiral crystal samples, the optical purity of the in situ induced cages was lower, which is consistent with the case of labile Fe_8Pd_6 due to fast dynamic configurations change in solution.⁶⁷ For the stable Co^{III} -cages, only neglectable CD signals were detected even after adding an excess amount of the *R/S*-BINOL inducers (Figure 3d,e). These results suggest significantly different stereochemical stability and dynamics between the Co^{II} and Co^{III} cages.

To better understand the stereochemical dynamic difference in these two states, the stability of the freshly prepared $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ was monitored at room temperature in CH_3CN solution following the removal of the *R*-BINOL inducer by washing the cages with Et_2O . As seen from Figure 4a–c, the Co^{II} cage underwent a fast racemization process with a rapid decay of CD signals; in contrast, the CD profile was maintained for $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ after 8 h. Furthermore, the reinduction experiments were carried out under the same conditions in attempts to convert $\Delta_8\text{-Co}_8\text{Pd}_6$ cages into $\Lambda_8\text{-Co}_8\text{Pd}_6$ cages (Figures S61–S63). Heating the CH_3CN solution of $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ at 60 °C for 2 h, the CD signal decreased dramatically due to the racemization of cage. When the opposite inducer *S*-BINOL was added, $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ further transformed into Λ -configuration, indicating that stereochemical interconversion could be easily implemented in the Co^{II} labile state. While in the stable form, $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ displayed strong stereostability even at 60 °C for 8 h at the cage assembly concentration, showing slight decrease of the CD signals. Further reinduction with *S*-BINOL inducer showcased much deterred conversion from $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ to its Λ -configuration.

Since the reduction process is faster than the racemization process in aqueous media,⁴⁶ we were also able to monitor the CD signal variation during valence transition from $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ to $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ in the presence of 100 equiv HQ in a $\text{CH}_3\text{CN}/\text{H}_2\text{O} = 1:4$ (v/v) mixture. It was noted that the first positive Cotton effect (~ 340 nm) in the long-wavelength remained positive with the signal slightly blue-shifted, while the second positive Cotton effect (~ 300 nm) gradually turned negative (Figure 4d). The overall CD pattern of $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ finally changed to that of $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$, closely matching those of as-prepared homochiral cages. These observations suggest that the chirality is preserved during the reduction processes; however, the Cotton effects exhibit pronounced variation, providing a signal feature suitable for spectroscopic analysis and tracking based on two pairs of mirror signals from $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$, $\Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$, $\Lambda_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ enantiomers, as well as two pairs of distinguishable signals from Δ_8 -

$\text{Co}^{\text{II}}_8\text{Pd}_6$, $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ and $\Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$, $\Lambda_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ isomers (Figure 5a).

To obtain detailed kinetic information on stereochemical difference for the interconversion of Δ_8 - and Λ_8 -cages in Co^{II} and Co^{III} states, the time- and temperature-dependent measurements of the CD intensity at 345 nm was performed and fitted to the equations of first-order reaction (Figures S64–S71).⁷⁹ The cages were placed in a sample cell with cryostat whose temperature was controlled within ± 0.1 °C. Due to the difference in the enantiomeric interconversion kinetics between two cages, lower temperatures (5–35 °C) and higher temperatures (55–65 °C) were applied for $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$, respectively, and the kinetic measurements were maintained in a constant concentration at 1.6×10^{-6} M. When one cage enantiomer transforms to an opposite one, the enantiomeric excess is reduced by two molecules; therefore, the real enantiomeric isomerization rate constant k_e is half of the observed racemization rate constant k .^{79,80} As listed in Tables S3 and S4, the k_e value for $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ at 35 °C was calculated to be $k_e^{\text{II}} = 4.02 \times 10^{-4} \text{ s}^{-1}$, which is an order of magnitude larger than $k_e^{\text{III}} = 1.60 \times 10^{-5} \text{ s}^{-1}$ for $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ at 55 °C, indicating much slower enantiomeric isomerization speed for the Co^{III} static state even at a higher temperature. With rate constants at variable temperatures in hand, the activation energy (E_a), can be obtained by fitting to Arrhenius equations. The fitted activation energy for the Co^{II} cage is $E_a^{\text{II}} = 9.43 \pm 0.78 \text{ kcal mol}^{-1}$, which is much lower than the E_a for the stable $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ ($E_a^{\text{III}} = 25.44 \pm 1.44 \text{ kcal mol}^{-1}$). Notably, the activation energy of $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ is below the limit (15–18 kcal mol^{-1}) for the widely held activation energy of racemization to enable enantioresolution,⁷⁹ which explains the reason why the chiral $\text{Co}^{\text{II}}_8\text{Pd}_6$ cage with high enantiopurity cannot be obtained through the in situ chiral-induction in solution. As depicted in Figure 5b, the racemization of the dynamic Δ_8 - and $\Lambda_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ cage via configurational conversion of the labile Co^{II} -centers can proceed easily with a low activation energy. In contrast, oxidation to Δ_8 - and $\Lambda_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ will rigidify the cage configuration, rendering higher energy barrier E_a to retard dynamic switch of chirality. Fitting the data by the Eyring–Polanyi equation gave thermodynamic parameters of Gibbs free energy of activation ($\Delta^\ddagger G$), enthalpy of activation ($\Delta^\ddagger H$), and entropy of activation ($\Delta^\ddagger S$). The chiral Co^{II} -cage exhibits a smaller $\Delta^\ddagger G$ with lower $\Delta^\ddagger H^{\text{II}}$ of $8.85 \pm 0.79 \text{ kcal mol}^{-1}$ and a more negative $\Delta^\ddagger S^{\text{II}}$ of $-44.01 \text{ cal mol}^{-1}$. In contrast, the chiral Co^{III} -cage shows a larger $\Delta^\ddagger G$ with increased $\Delta^\ddagger H^{\text{III}}$ of $24.88 \pm 1.45 \text{ kcal mol}^{-1}$ and $\Delta^\ddagger S^{\text{III}}$ of $-3.37 \text{ cal mol}^{-1}$. The positive $\Delta^\ddagger H$ and negative $\Delta^\ddagger S$ values results in positive $\Delta^\ddagger G$, indicating the racemization is endothermic and could not proceed spontaneously.

The mechanism of the stereochemical transformation through configurational interconversion based on the lability of Co^{II} -centers may be postulated from the following experimental results. First, the absolute configuration just changes gradually during the racemization process, which could be further reinduced to the opposite one (Figures 4a and S61), and both $\Delta_8\text{-Co}^{\text{II}}_8\text{Pd}_6$ and $\Delta_8\text{-Co}^{\text{III}}_8\text{Pd}_6$ cages preserve their absolute configurations during valence transition (Figure 4d). Second, the anion-induced mesomerization of $\Delta_4/\Lambda_4\text{-Co}^{\text{II}}_8\text{Pd}_6$ happened,⁴⁶ leading to the formation of *meso*-configurations in one cage. In the presence of excess KPF₆ during crystallization, we obtained the single-crystal of *meso*- $\Delta_4/\Lambda_4\text{-Co}^{\text{II}}_8\text{Pd}_6\text{-PF}_6$ (Figures 4g,h and S31). In contrast, the

Co^{III} cage was unaffected by KPF₆ and retained the homochiral cage configuration under similar inducing conditions (Figure 4e,f). These observations suggest a plausible stereochemical transformation route via a *meso*-configuration forming step (Figure 5b), benefiting from the lability of the Co^{II}-centers and the stereochemical dynamics of Co^{II}-cages.

Based on above-described reversible redox and stereo-inversion processes, a chiral optical logic gate may be proposed as depicted in Figure 5a. The locking states are presented by Λ_8 -Co^{III}₈Pd₆ (S1) and Δ_8 -Co^{III}₈Pd₆ (S2) enantiomers, while the unlocking states are presented by Λ_8 -Co^{II}₈Pd₆ (L1) and Δ_8 -Co^{II}₈Pd₆ (L2) enantiomers, which display characteristic but distinct signal features of mirror or differential Cotton effects. The turn-on and turn-off switch can be implemented by reduction and oxidation processes between labile Co^{II}₈Pd₆ and static Co^{III}₈Pd₆ cages, while the chirality switch can be regulated by enantioselective recognition of the guests between Δ_8 -Co^{II}₈Pd₆ and Λ_8 -Co^{II}₈Pd₆ cages via chiral induction. Following these reversible processes, on the one hand, the valence transition and stereoconfiguration transformation can be tracked, and on the other hand, different inputs of the chiral guests can lead to opposite CD signal outputs.

CONCLUSIONS

In summary, we have demonstrated a redox-switchable protocol to control the stereochemical stability of heterometallic Co₈Pd₆ cages, offering a convenient way to manipulate the structural and dynamic modulations among different valences and chirality. The introduction of redox-active Co^{II}/Co^{III}-centers allows for facile transition between labile and static states via charge isomerization, facilitated by mild redox conditions. The stereolabile Co^{II}-centers endow Λ_8/Δ_8 -Co^{II}₈Pd₆ cages with stereochemical dynamics to exhibit tunable enantiomerism with stereoconfigurations easily induced and altered by enantioselectors, possibly through a mesoisomerization step. The oxidation of Λ_8/Δ_8 -Co^{II}₈Pd₆ to Λ_8/Δ_8 -Co^{III}₈Pd₆ brings in a significant cage rigidity and locks the cage stereoconfigurations to prevent racemization. Moreover, intriguing Cotton effects are presented in four distinguishable charge or stereo isomers. Therefore, such redox- and chirality-switchable systems hold considerable promise for applications requiring precise control over structural stability to lock or unlock the supramolecular functions, achieving stereoselective recognition and transformation detectable by spectroscopic analysis. This work may inspire future design of chiral molecular devices to combine the high stereostability with tunable dynamics for functional switches.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c06102>.

Experimental procedures, crystallographic data, and additional NMR and ESI-MS spectra (PDF)

Accession Codes

CCDC 2206002–2206003, 2206005–2206009, and 2348552 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Chakraborty, R.; Mukherjee, P. S.; Stang, P. J. Supramolecular Coordination: Self-Assembly of Finite Two- and Three-Dimensional Ensembles. *Chem. Rev.* **2011**, *111*, 6810–6918.
- (2) Cook, T. R.; Zheng, Y.-R.; Stang, P. J. Metal-Organic Frameworks and Self-Assembled Supramolecular Coordination Complexes: Comparing and Contrasting the Design, Synthesis, and Functionality of Metal-Organic Materials. *Chem. Rev.* **2013**, *113*, 734–777.
- (3) Pan, M.; Wu, K.; Zhang, J.-H.; Su, C.-Y. Chiral Metal-Organic Cages/Containers (MOCs): From Structural and Stereochemical Design to Applications. *Coord. Chem. Rev.* **2019**, *378*, 333–349.
- (4) Han, M.; Engelhard, D. M.; Clever, G. H. Self-Assembled Coordination Cages Based on Banana-Shaped Ligands. *Chem. Soc. Rev.* **2014**, *43*, 1848–1860.
- (5) Ronson, T. K.; Zarra, S.; Black, S. P.; Nitschke, J. R. Metal-Organic Container Molecules Through Subcomponent Self-Assembly. *Chem. Commun.* **2013**, *49*, 2476–2490.
- (6) Brown, C. J.; Toste, F. D.; Bergman, R. G.; Raymond, K. N. Supramolecular Catalysis in Metal-Ligand Cluster Hosts. *Chem. Rev.* **2015**, *115*, 3012–3035.
- (7) Koblenz, T. S.; Wassenaar, J.; Reek, J. N. H. Reactivity within a Confined Self-Assembled Nanospace. *Chem. Soc. Rev.* **2008**, *37*, 247–262.
- (8) Yoshizawa, M.; Tamura, M.; Fujita, M. Diels-Alder in Aqueous Molecular Hosts: Unusual Regioselectivity and Efficient Catalysis. *Science* **2006**, *312*, 251–254.
- (9) Yoshizawa, M.; Klosterman, J. K.; Fujita, M. Functional Molecular Flasks: New Properties and Reactions within Discrete, Self-Assembled Hosts. *Angew. Chem., Int. Ed.* **2009**, *48*, 3418–3438.
- (10) Martí-Centelles, V.; Lawrence, A. L.; Lusby, P. J. High Activity and Efficient Turnover by a Simple, Self-Assembled “Artificial Diels-Alderase”. *J. Am. Chem. Soc.* **2018**, *140*, 2862–2868.
- (11) Wang, Q. Q.; Gonell, S.; Leenders, S. H.; Durr, M.; Ivanovic-Burmazovic, I.; Reek, J. N. Self-Assembled Nanospheres with Multiple Endohedral Binding Sites Pre-organize Catalysts and Substrates for Highly Efficient Reactions. *Nat. Chem.* **2016**, *8*, 225–230.
- (12) Wang, J.-S.; Wu, K.; Yin, C.; Li, K.; Huang, Y.; Ruan, J.; Feng, X.; Hu, P.; Su, C.-Y. Cage-Confined Photocatalysis for Wide-Scope Unusually Selective [2 + 2] Cycloaddition Through Visible-Light Triplet Sensitization. *Nat. Commun.* **2020**, *11*, 4675.
- (13) Gou, X.-X.; Liu, T.; Wang, Y.-Y.; Han, Y.-F. Ultrastable and Highly Catalytically Active N-Heterocyclic-Carbene-Stabilized Gold Nanoparticles in Confined Spaces. *Angew. Chem., Int. Ed.* **2020**, *59*, 16683–16689.
- (14) Grommet, A. B.; Feller, M.; Klajn, R. Chemical Reactivity Under Nanoconfinement. *Nat. Nanotechnol.* **2020**, *15*, 256–271.
- (15) Howlader, P.; Das, P.; Zangrando, E.; Mukherjee, P. S. Urea-Functionalized Self-Assembled Molecular Prism for Heterogeneous Catalysis in Water. *J. Am. Chem. Soc.* **2016**, *138*, 1668–1676.
- (16) Ward, M. D.; Hunter, C. A.; Williams, N. H. Coordination Cages Based on Bis(pyrazolylpyridine) Ligands: Structures, Dynamic Behavior, Guest Binding, and Catalysis. *Acc. Chem. Res.* **2018**, *51*, 2073–2082.
- (17) Hao, Y.; Lu, Y.-L.; Jiao, Z.; Su, C.-Y. Photocatalysis Meets Confinement: An Emerging Opportunity for Photoinduced Organic Transformations. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e202317808.
- (18) Chen, J.; Wu, X.; Huang, S.; Yang, J.; Lu, Y.-L.; Jiao, Z.; Su, C.-Y. Catalytic Enantioselective Cycloaddition Transformation of Tricyclic Arenes Enabled by a Dual-Role Chiral Cage-Reactor. *ACS Catal.* **2024**, *14*, 3733–3741.
- (19) Yang, Y.; Li, H.; Shi, Y.; Wu, Y.; Jing, X.; Duan, C. Modifying the Oxidative Potentials of Imines in a Dye Loaded Capsule for Photocatalytic Cyclization with Hydrogen Evolution. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e202319605.
- (20) Ham, R.; Nielsen, C. J.; Pullen, S.; Reek, J. N. H. Supramolecular Coordination Cages for Artificial Photosynthesis and Synthetic Photocatalysis. *Chem. Rev.* **2023**, *123*, 5225–5261.
- (21) Zhang, D.; Ronson, T. K.; Zou, Y.-Q.; Nitschke, J. R. Metal-Organic Cages for Molecular Separations. *Nat. Rev. Chem.* **2021**, *5*, 168–182.
- (22) Hasell, T.; Cooper, A. I. Porous Organic Cages: Soluble, Modular and Molecular Pores. *Nat. Rev. Mater.* **2016**, *1*, 16053.
- (23) Li, X.-Z.; Zhou, L.-P.; Yan, L.-L.; Dong, Y.-M.; Bai, Z.-L.; Sun, X.-Q.; Diwu, J.; Wang, S.; Bünzli, J.-C.; Sun, Q.-F. A Supramolecular Lanthanide Separation Approach Based on Multivalent Cooperative Enhancement of Metal Ion Selectivity. *Nat. Commun.* **2018**, *9*, 547.
- (24) García-Simón, C.; Costas, M.; Ribas, X. Metallosupramolecular Receptors for Fullerene Binding and Release. *Chem. Soc. Rev.* **2016**, *45*, 40–62.
- (25) Wang, L. J.; Bai, S.; Han, Y. F. Water-Soluble Self-Assembled Cage with Triangular Metal-Metal-Bonded Units Enabling the Sequential Selective Separation of Alkanes and Isomeric Molecules. *J. Am. Chem. Soc.* **2022**, *144*, 16191–16198.
- (26) Zhu, C. F.; Yang, K. K.; Wang, H. Z.; Fang, Y.; Feng, L.; Zhang, J. Q.; Xiao, Z. F.; Wu, X.; Li, Y. G.; Fu, Y. M.; Zhang, W. C.; Wang, K. Y.; Zhou, H. C. Enantioseparation in Hierarchically Porous Assemblies of Homochiral Cages. *ACS Cent. Sci.* **2022**, *8*, 562–570.
- (27) Galan, A.; Ballester, P. Stabilization of Reactive Species by Supramolecular Encapsulation. *Chem. Soc. Rev.* **2016**, *45*, 1720–1737.
- (28) Mal, P.; Breiner, B.; Rissanen, K.; Nitschke, J. R. White Phosphorus Is Air-Stable Within a Self-Assembled Tetrahedral Capsule. *Science* **2009**, *324*, 1697–1699.
- (29) Li, K.; Zhang, L. Y.; Yan, C.; Wei, S. C.; Pan, M.; Zhang, L.; Su, C. Y. Stepwise Assembly of Pd₆(RuL₃)₈ Nanoscale Rhombododecahedral Metal-Organic Cages via Metalloligand Strategy for Guest Trapping and Protection. *J. Am. Chem. Soc.* **2014**, *136*, 4456–4459.
- (30) Yamashina, M.; Sei, Y.; Akita, M.; Yoshizawa, M. Safe Storage of Radical Initiators within a Polyaromatic Nanocapsule. *Nat. Commun.* **2014**, *5*, 4662.
- (31) Cook, T. R.; Vajpayee, V.; Lee, M. H.; Stang, P. J.; Chi, K.-W. Biomedical and Biochemical Applications of Self-Assembled Metal-lacycles and Metallacages. *Acc. Chem. Res.* **2013**, *46*, 2464–2474.
- (32) Dong, J.; Pan, Y.; Wang, H.; Yang, K.; Liu, L.; Qiao, Z.; Yuan, Y. D.; Peh, S. B.; Zhang, J.; Shi, L.; Liang, H.; Han, Y.; Li, X.; Jiang, J.; Liu, B.; Zhao, D. Self-Assembly of Highly Stable Zirconium(IV) Coordination Cages with Aggregation Induced Emission Molecular Rotors for Live-Cell Imaging. *Angew. Chem., Int. Ed.* **2020**, *59*, 10151–10159.
- (33) Samanta, S. K.; Moncelet, D.; Briken, V.; Isaacs, L. Metal-Organic Polyhedron Capped with Cucurbit[8]uril Delivers Doxorubicin to Cancer Cells. *J. Am. Chem. Soc.* **2016**, *138*, 14488–14496.
- (34) Zheng, Y.-R.; Suntharalingam, K.; Johnstone, T. C.; Lippard, S. J. Encapsulation of Pt(IV) Prodrugs within a Pt(II) Cage for Drug Delivery. *Chem. Sci.* **2015**, *6*, 1189–1193.
- (35) Lewis, J. E. M.; Gavey, E. L.; Cameron, S. A.; Crowley, J. D. Stimuli-Responsive Pd₂L₄ Metallosupramolecular Cages: Towards Targeted Cisplatin Drug Delivery. *Chem. Sci.* **2012**, *3*, 778–784.
- (36) Percástegui, E. G.; Ronson, T. K.; Nitschke, J. R. Design and Applications of Water-Soluble Coordination Cages. *Chem. Rev.* **2020**, *120*, 13480–13544.

- (37) Ahmad, N.; Younus, H. A.; Chughtai, A. H.; Verpoort, F. Metal-Organic Molecular Cages: Applications of Biochemical Implications. *Chem. Soc. Rev.* **2015**, *44*, 9–25.
- (38) Wang, Y. P.; Wu, K.; Pan, M.; Li, K.; Mo, J. T.; Duan, X. H.; He, H. Z.; Shen, J.; Su, C. Y. One-/Two-Photon Excited Cell Membrane Imaging and Tracking by a Photoactive Nanocage. *ACS Appl. Mater. Interfaces* **2020**, *12*, 35873–35881.
- (39) Wang, Y.-P.; Zhang, Y.; Duan, X.-H.; Mao, J.-J.; Pan, M.; Shen, J.; Su, C.-Y. Recent progress in metal-organic cages for biomedical application: Highlighted research during 2018–2023. *Coord. Chem. Rev.* **2024**, *501*, 215570.
- (40) Erbas-Cakmak, S.; Leigh, D. A.; McTernan, C. T.; Nussbaumer, A. L. Artificial Molecular Machines. *Chem. Rev.* **2015**, *115*, 10081–10206.
- (41) Park, J. S.; Sessler, J. L. Tetrathiafulvalene (TTF)-Annulated Calix[4]pyrroles: Chemically Switchable Systems with Encodable Allosteric Recognition and Logic Gate Functions. *Acc. Chem. Res.* **2018**, *51*, 2400–2410.
- (42) Yazaki, K.; Noda, S.; Tanaka, Y.; Sei, Y.; Akita, M.; Yoshizawa, M. An M_2L_4 Molecular Capsule with a Redox Switchable Polyradical Shell. *Angew. Chem., Int. Ed.* **2016**, *55*, 15031–15034.
- (43) Ube, H.; Endo, K.; Sato, H.; Shionoya, M. Synthesis of Hetero-multinuclear Metal Complexes by Site-Selective Redox Switching and Transmetalation on a Homo-multinuclear Complex. *J. Am. Chem. Soc.* **2019**, *141*, 10384–10389.
- (44) Li, R.-J.; Holstein, J. J.; Hiller, W. G.; Andréasson, J.; Clever, G. H. Mechanistic Interplay between Light Switching and Guest Binding in Photochromic $[Pd_2Dithienylethene_4]$ Coordination Cages. *J. Am. Chem. Soc.* **2019**, *141*, 2097–2103.
- (45) Dumartin, M.; Lipke, M. C.; Stoddart, J. F. A Redox-Switchable Molecular Zipper. *J. Am. Chem. Soc.* **2019**, *141*, 18308–18317.
- (46) Huang, Y.-H.; Lu, Y.-L.; Zhang, X.-D.; Liu, C.-H.; Ruan, J.; Qin, Y.-H.; Cao, Z.-M.; Jiang, J.-J.; Xu, H.-S.; Su, C.-Y. Dynamic Stereochemistry of M_3Pd_6 Supramolecular Cages Based on Metal-Center Liability for Differential Chiral Induction, Resolution, and Recognition. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e202315053.
- (47) Lu, Y.-L.; Zhang, X.-D.; Qin, Y.-H.; Song, J.-Q.; Huang, Y.-H.; Liu, C.-H.; Chen, J.-J.; Xu, H.-S.; Pan, M.; Su, C.-Y. A robust protein-mimicking metallo-amine cage showing proton-driven allostery with water as the effector. *Chem* **2023**, *9*, 2144–2160.
- (48) Xie, T. Z.; Liao, S. Y.; Guo, K.; Lu, X.; Dong, X.; Huang, M.; Moorefield, C. N.; Cheng, S. Z.; Liu, X.; Wesdemiotis, C.; Newkome, G. R. Construction of a Highly Symmetric Nanosphere via a One-Pot Reaction of a Tris(terpyridine) Ligand with Ru(II). *J. Am. Chem. Soc.* **2014**, *136*, 8165–8168.
- (49) Wu, K.; Li, K.; Hou, Y. J.; Pan, M.; Zhang, L. Y.; Chen, L.; Su, C. Y. Homochiral D_4 -Symmetric Metal-Organic Cages from Stereogenic Ru(II) Metalloligands for Effective Enantioseparation of Atropisomeric Molecules. *Nat. Commun.* **2016**, *7*, 10487.
- (50) Burke, M. J.; Nichol, G. S.; Lusby, P. J. Orthogonal Selection and Fixing of Coordination Self-Assembly Pathways for Robust Metallo-organic Ensemble Construction. *J. Am. Chem. Soc.* **2016**, *138*, 9308–9315.
- (51) Su, P.; Zhang, W.; Guo, C.; Liu, H.; Xiong, C.; Tang, R.; He, C.; Chen, Z.; Yu, X.; Wang, H.; Li, X. Constructing Ultrastable Metallo-Cages via In Situ Deprotonation/Oxidation of Dynamic Supramolecular Assemblies. *J. Am. Chem. Soc.* **2023**, *145*, 18607–18622.
- (52) Holmstrom, K. M.; Finkel, T. Cellular mechanisms and physiological consequences of redox-dependent signalling. *Nat. Rev. Mol. Cell Biol.* **2014**, *15*, 411–421.
- (53) Pochapsky, S. S.; Dang, M.; OuYang, B.; Simorellis, A. K.; Pochapsky, T. C. Redox-dependent dynamics in cytochrome P450cam. *Biochemistry* **2009**, *48*, 4254–4261.
- (54) Zhong, F.; Lisi, G. P.; Collins, D. P.; Dawson, J. H.; Pletneva, E. V. Redox-dependent stability, protonation, and reactivity of cysteine-bound heme proteins. *Proc. Natl. Acad. Sci. U.S.A.* **2014**, *111*, E306–E315.
- (55) Hall, A.; Parsonage, D.; Horita, D.; Karplus, P. A.; Poole, L. B.; Barbar, E. Redox-dependent dynamics of a dual thioredoxin fold protein: evolution of specialized folds. *Biochemistry* **2009**, *48*, 5984–5993.
- (56) Skeens, E.; Gadzik-Shea, M.; Shah, D.; Bhandari, V.; Schweppe, D. K.; Berlow, R. B.; Lisi, G. P. Redox-dependent structure and dynamics of macrophage migration inhibitory factor reveal sites of latent allostery. *Structure* **2022**, *30*, 840–850.e6.
- (57) Su, P.; Wei, B.; Guo, C.; Hu, Y.; Tang, R.; Zhang, S.; He, C.; Lin, J.; Yu, X.; Chen, Z.; Li, H.; Wang, H.; Li, X. Metallo-Supramolecular Hexagonal Wreath with Four Switchable States Based on a pH-Responsive Tridentate Ligand. *J. Am. Chem. Soc.* **2023**, *145*, 3131–3145.
- (58) Szalóki, G.; Croué, V.; Carré, V.; Aubriet, F.; Alévêque, O.; Levillain, E.; Allain, M.; Aragón, J.; Ortí, E.; Goeb, S.; Sallé, M. Controlling the Host-Guest Interaction Mode through a Redox Stimulus. *Angew. Chem., Int. Ed.* **2017**, *56*, 16272–16276.
- (59) Lee, B.; Go, B.; Jung, B.; Park, J. Unlocking High Porosity: Post-Synthetic Solvothermal Treatment of Cu-Paddlewheel Based Metal-Organic Cages. *Small* **2024**, *20*, No. e2308393.
- (60) Huang, Y.-H.; Lu, Y.-L.; Ruan, J.; Zheng, S.-P.; Zhang, X.-D.; Liu, C.-H.; Qin, Y.-H.; Cao, Z.-M.; Jiao, Z.; Xu, H.-S.; Su, C.-Y. Dynamic Metallosupramolecular Cages Containing 12 Adaptable Pockets for High-Order Guest Binding Beyond Biomimicry. *J. Am. Chem. Soc.* **2023**, *145*, 23361–23371.
- (61) Ronson, T. K.; Giri, C.; Beyeh, N. K.; Minkinen, A.; Topic, F.; Holstein, J. J.; Rissanen, K.; Nitschke, J. R. Size-selective encapsulation of hydrophobic guests by self-assembled M_4L_6 cobalt and nickel cages. *Chem.—Eur. J.* **2013**, *19*, 3374–3382.
- (62) Xue, W.; Ronson, T. K.; Lu, Z.; Nitschke, J. R. Solvent Drives Switching between Lambda and Delta Metal Center Stereochemistry of M_3L_6 Cubic Cages. *J. Am. Chem. Soc.* **2022**, *144*, 6136–6142.
- (63) Riddell, I. A.; Smulders, M. M.; Clegg, J. K.; Hristova, Y. R.; Breiner, B.; Thoburn, J. D.; Nitschke, J. R. Anion-Induced Reconstitution of a Self-Assembling System to Express a Chloride-Binding $Co_{10}L_{15}$ Pentagonal Prism. *Nat. Chem.* **2012**, *4*, 751–756.
- (64) Symmers, P. R.; Burke, M. J.; August, D. P.; Thomson, P. I. T.; Nichol, G. S.; Warren, M. R.; Campbell, C. J.; Lusby, P. J. Non-equilibrium cobalt(III) “click” capsules. *Chem. Sci.* **2015**, *6*, 756–760.
- (65) Björemark, P. M.; Jönsson, J.; Håkansson, M. Absolute Asymmetric Synthesis: Viedma Ripening of $[Co(bpy)_3]^{2+}$ and Solvent-Free Oxidation to $[Co(bpy)_3]^{3+}$. *Chem.—Eur. J.* **2015**, *21*, 10630–10633.
- (66) Rodger, A.; Nordén, B.; Rodger, P. M.; Bates, Paula, J. DNA as a Catalyst and Catalytic Template for the Racemisation of Metal Tris-Phenanthroline Complexes. *Eur. J. Inorg. Chem.* **2002**, *2002*, 49–53.
- (67) Hou, Y. J.; Wu, K.; Wei, Z. W.; Li, K.; Lu, Y. L.; Zhu, C. Y.; Wang, J. S.; Pan, M.; Jiang, J. J.; Li, G. Q.; Su, C. Y. Design and Enantioresolution of Homochiral Fe(II)-Pd(II) Coordination Cages from Stereolabile Metalloligands: Stereochemical Stability and Enantioselective Separation. *J. Am. Chem. Soc.* **2018**, *140*, 18183–18191.
- (68) Muntener, T.; Joss, D.; Haussinger, D.; Hiller, S. Pseudocontact Shifts in Biomolecular NMR Spectroscopy. *Chem. Rev.* **2022**, *122*, 9422–9467.
- (69) Miao, Q.; Nitsche, C.; Orton, H.; Overhand, M.; Otting, G.; Ubbink, M. Paramagnetic Chemical Probes for Studying Biological Macromolecules. *Chem. Rev.* **2022**, *122*, 9571–9642.
- (70) Lu, Y. L.; Song, J. Q.; Qin, Y. H.; Guo, J.; Huang, Y. H.; Zhang, X. D.; Pan, M.; Su, C. Y. A Redox-Active Supramolecular Fe_4L_6 Cage Based on Organic Vertices with Acid-Base-Dependent Charge Tunability for Dehydrogenation Catalysis. *J. Am. Chem. Soc.* **2022**, *144*, 8778–8788.
- (71) Prikhod'ko, A. I.; Sauvage, J.-P. Passing Two Strings through the Same Ring Using an Octahedral Metal Center as Template: A New Synthesis of [3]Rotaxanes. *J. Am. Chem. Soc.* **2009**, *131*, 6794–6807.
- (72) Leigh, D. A.; Lusby, P. J.; McBurney, R. T.; Morelli, A.; Slawin, A. M. Z.; Thomson, A. R.; Walker, D. B. Getting Harder: Cobalt(III)-

Template Synthesis of Catenanes and Rotaxanes. *J. Am. Chem. Soc.* **2009**, *131*, 3762–3771.

(73) Murase, T.; Sato, S.; Fujita, M. Switching the interior hydrophobicity of a self-assembled spherical complex through the photoisomerization of confined azobenzene chromophores. *Angew. Chem., Int. Ed.* **2007**, *46*, 5133–5136.

(74) Yang, C.; Nikiforidis, G.; Park, J. Y.; Choi, J.; Luo, Y.; Zhang, L.; Wang, S.-C.; Chan, Y.-T.; Lim, J.; Hou, Z.; Baik, M.-H.; Lee, Y.; Byon, H. R. Designing Redox-Stable Cobalt-Polypyridyl Complexes for Redox Flow Batteries: Spin-Crossover Delocalizes Excess Charge. *Adv. Energy Mater.* **2018**, *8*, 1702897.

(75) Cho, I.; Koshika, M.; Wagner, P.; Koumura, N.; Innis, P. C.; Mori, S.; Mozer, A. J. Exploiting Intermolecular Interactions between Alkyl-Functionalized Redox-Active Molecule Pairs to Enhance Interfacial Electron Transfer. *J. Am. Chem. Soc.* **2018**, *140*, 13935–13944.

(76) Feldt, S. M.; Gibson, E. A.; Gabrielsson, E.; Sun, L.; Boschloo, G.; Hagfeldt, A. Design of Organic Dyes and Cobalt Polypyridine Redox Mediators for High-Efficiency Dye-Sensitized Solar Cells. *J. Am. Chem. Soc.* **2010**, *132*, 16714–16724.

(77) Mtshali, Z.; Conradie, J. Tris(polypyridine)nickel(II) complexes: Synthesis, DFT and electrochemistry. *Inorg. Chim. Acta* **2023**, *549*, 121422.

(78) Guo, J.; Xu, Y.-W.; Li, K.; Xiao, L.-M.; Chen, S.; Wu, K.; Chen, X.-D.; Fan, Y.-Z.; Liu, J.-M.; Su, C.-Y. Regio- and Enantioselective Photodimerization within the Confined Space of a Homochiral Ruthenium/Palladium Heterometallic Coordination Cage. *Angew. Chem., Int. Ed.* **2017**, *56*, 3852–3856.

(79) Harada, N.; Saito, A.; Koumura, N.; Roe, D. C.; Jager, W. F.; Zijlstra, R. W. J.; de Lange, B.; Feringa, B. L. Chemistry of Unique Chiral Olefins. 2. Unexpected Thermal Racemization of *cis*-1,1',2,2',3,3',4,4'-Octahydro-4,4'-biphenanthrylidene. *J. Am. Chem. Soc.* **1997**, *119*, 7249–7255.

(80) Ravat, P. Carbo[n]helicenes Restricted to Enantiomerize: An Insight into the Design Process of Configurationally Stable Functional Chiral PAHs. *Chem.—Eur. J.* **2021**, *27*, 3957–3967.